Research Portfolio

Tianmai Ryan Xiao

BASIS International School Shenzhen; Shenzhen Medical Academy of Research and Translation; Peking University Shenzhen Graduate School; University of Florida Student Science Training Program; Beijing Zhongguancun Academy

Research Directions

Molecular biology, Neuroscience, Neurogenetics, Computational psychology

Research Overview

My overall research interest lies in understanding the biology behind the brain. More specifically, I am interested in how the brain changes as we learn. My ambitions in this subject field consist of discovering novel neural mechanisms that can enhance the learning process, help mitigate the effects of learning disabilities, and/or discover broader educational psychology methods that may enhance learning. Through exploring multiple research fields such as molecular biology, neuroscience, and neurogenetics, and through the papers, reviews, and individual research I have conducted myself, I have learned and grown as an aspiring scientist and educator. Simultaneously, I have watched many of the neural mechanisms that I have been explored working in big and small ways around me. As I explored this new material and became more proficient within it, I have thus developed an intense curiosity towards what neural mechanisms could be happening inside the brain.

Recently, I have been designing a computational psychology research project investigating human interactions with Artificial Intelligence and Large-Language Models. This serves for the purpose of familiarizing myself with experimental design and conducting independent research, as well as familiarizing myself with vibe-coding. This project is ongoing.

Collaborative Research

Mapping the biophysics of voltage-gated sodium channels $Na_v1.7$ (June to July, 2024, 4.5 weeks)

Research Assistant, Dr. Xiaojing Pan Laboratory, Shenzhen Medical Academy of Research and Technology

Research Context

Sodium voltage channels are common membrane receptors found in many cell types. For example, they can be found along the axon of a neuron. Because their prominent role in cellular communication, they have been targeted by many drugs treating disorders such as epilepsy or pain symptoms. However, the binding mode of many $Na_v1.7$ remains unknown. This lab found multiple structural components of the receptor that could better inform drug binding and help future studies develop 3D models of the receptor under cryo-EM technology.

My Roles

I shadowed graduate students in their work. Observed the development of plasmids, amplification, and transformation to *E. coli*. Observed the extraction and purification of the proteins of interest. Familiarized myself with common biology equipment and techniques like gel electrophoresis, centrifuges, and the DNA extraction protocol from bacterial cells.

I Developed a series of short informational videos on lab equipment safety and usage, and posted on the website, BiliBili.

Skills Developed

- PCR and gel electrophoresis
- DNA extraction

Reflection

Through this project, I came to understand some basic procedures that almost all biology labs have to undergo to obtain genetic material or purified proteins. I realized how science is not always directly about the experiment but about the preparation process for the main investigative purposes instead.

Studying the role of G protein receptors in Autistic Spectrum Disorder (July to August 2025, 7 weeks)

Research Assistant, Dr. Bo Zhang Laboratory, Peking University Graduate School at Shenzhen

Research Context

G protein-coupled receptors are membrane receptors found in many neurons in the cell body membrane. Studies have shown that they play an important role in neuronal communication. Autistic Spectrum Disorder is a neurodevelopmental disorder associated commonly with social impairment, language impairment, or other symptoms. However, the link between neural networks that can cause these symptoms in Autistic Spectrum Disorder is not well known. Our lab hypothesized that mutations in certain G protein receptors could disrupt neuronal communication pathways that in turn lead to the phenotypical symptoms exhibited by autistic individuals.

My Roles

I assisted a graduate student Heping Zhang with her research. I conducted behavioral experiments such as open-field tests, tail suspension tests and three chamber tests. I observed autopsy of mouse brains, and the fixing, slicing, and imaging. I observed construction of plasmids and amplified them with *E. coli* and validated template DNA through PCR. I observed qPCR. Observed the injection of viral vectors containing genetic mutations into the mouse brain. I observed neuronal culture and biophysiology experiments of neurons. I extracted mouse brains as practice. Attended weekly journal club meetings. Every day, I took detailed lab logs.

Skills Developed

- PCR
- Plasmid amplification with *E. coli*
- Fixed mouse brain tissue extraction
- Behavioral tests (OFT, TCT, TST)

Reflection

My experience in Dr. Zhang's lab pinpointed my research interest in neuroscience. I was really fascinated by how behavioral phenotypes, statistical analysis and protein imaging could all confirm the mice’s social impairment-causing mutation. This made me see the link between molecular-level protein expression and organism-level phenotypes, instead of working at one level of breadth as I had done previously.

Mentor-guided Independent Research

Studying the toxic effects of Alzheimer's risk variant of *CASP8* using cellular models (June to July 2026, 7 weeks)

Mentors: Denise Romero and Rodrigo Tomas

Dr. Lien Nguyen Laboratory, Student Science Training Program, University of Florida

Research Question

Can differentiated SH-SY5Y cells transduced with *CASP8*-AD-R1 model Alzheimer's Disease neuropathological changes?

Background and Significance

Alzheimer's disease is the most common form of dementia, but its genetic causes are not fully understood. Our lab has discovered a repeat expansion mutation on the gene *CASP8* (*CASP8*-AD-R1) that has been linked significantly to Alzheimer's risk. Although this lab has used mammalian cell models to demonstrate that this variant can cause an increase in pathogenic proteins (such as pTau) related to Alzheimer's (Nguyen et al., 2025 & Phuong and Tomas et al., 2025), it had yet to use a model that resembles the human brain even more to understand if this phenomenon occurs within patients. Using this refined model, this lab could begin targeting this mechanism for Alzheimer's disease therapeutics.

Hypothesis

That the expression of *CASP8*-AD-R1 using an adeno-associated virus (AAV) approach in differentiated human neuroblastoma (SH-SY5Y) cells will lead to increased pTau levels and contribute to cytotoxicity.

Methods

The cellular model of SH-SY5Y cells were used to observe for pathological impacts of *CASP8*-AD-R1. This involved differentiating the SH-SY5Y to become mature neuron-like cells, and in the process transducing them with an AAV, then performing immunofluorescence staining (IF) on the proteins of interest.

Differentiation occurs for 7 days, with days 1, 4, and 6 involving treatments with retinoic acid and 1% Fetal Bovine Serum. Viral transduction occurs at day 7. Cells mature for 14 more days before they are fixed and IF stained.

Results

Brightfield images showed marks of initial differentiation. In general, cells were observed to have grown in axonal length, formed clusters, and have increased branching. This indicated that the differentiation protocol was successful as morphological changes that are associated with mature neurons were observed. To confirm these changes happened at the molecular level, there was staining for mature neuronal markers.

Next, to investigate if C8-AD-R1-expressing cells had increased pTau, qualitative data was collected. Observations suggested that there could be a relationship between the risk variant and pTau in this scenario as there is much more colocalization that can be observed in the AD-R1 group compared to the control. However, there was not sufficient results to conduct statistical analysis; therefore, these results are merely suggestive.

Skills Developed

- Cell culture (passaging, changing media, cell-counting, AAV transduction)
- Immunofluorescence staining (fixed cells and fixed tissue)
- Confocal microscopy

- Statistical analysis of fluorescent imaging

Discussion

This study suggests that SH-SY5Y cells could be suitable models for further investigation of AD. A 21-day differentiation protocol utilizing RA can successfully lead them to grow into mature neuron-like cells, making them effective *in vitro* models for neuroscience research. Other research topics, such as other repeat expansion mutations related to AD or neurological disorders, could also utilize SH-SY5Y models.

Because we utilized a cellular model, we could only effectively investigate intracellular mechanisms through IF staining. Future experimentation should investigate the link between *CASP8*-AD-R1 and extracellular Alzheimer's pathology. Further solidification of findings could encompass different staining methods, such as immunohistochemistry (IHC) or RNA Fluorescence In Situ Hybridization (RNA-FISH) staining. Future directions include investigating extracellular pathology, different neuronal types, and different brain cell types.

Persistence and Change in Requested AI Functions: An Observational Study of WildChat Conversations (September 2026 to present ongoing)

Mentor: Dr. Chen Hu, Beijing Zhongguancun Academy

Research Question

How is the function users initially request of an AI associated with persistence or change in the tasks or judgement they delegate during subsequent conversation?

Background and Significance

Users can ask AI to perform many functions, from information-seeking to text editing to evaluating and recommending decisions. We wished to investigate how these requests develop

throughout a conversation, if users tended to continue asking for the same type of assistance, or if they changed the topic of the conversation. Understanding these patterns could inform the design of AI systems that better support users as their needs evolve, as well as provide insight into how we cognitively are adapting to this new multi-capable tool.

Methods

This project uses an observational analysis of conversations from WildChat, a dataset of user–AI interactions. An initial pilot of 30 conversations containing 64 user follow-ups supported the development and refinement of an annotation framework.

The framework records task type, delegated judgment, follow-up actions, and explicit evaluative feedback. Secondary measures describe whether users develop existing content or introduce new directions, and whether developed content originated with the user, AI, or both. User satisfaction is either taken from explicit feedback or assessed by an external LLM to infer it.

A formal sample of 300 English-language GPT-3.5-turbo conversations was selected through reproducible random ordering and eligibility screening, with 922 total user follow-ups. LLM-assisted annotation uses the conversation history available through each target message, excluding later responses. Each substantive follow-up is compared with both the initial request and the preceding substantive request to identify persistence and change.

Results

As this project is still ongoing, we have yet to obtain and analyze results.

Skills Developed

- Independent research design and operationalization of behavioral constructs
- Development and refinement of qualitative annotation codebooks

- Data processing, sampling, and validation
- LLM-assisted annotation and evidence-based coding
- Vibe-coding-based modeling

References

Nguyen, L., Ajredini, R., Guo, S., Romano, L. E. L., Tomas, R. F., Bell, L. R., Ranum, P. T., Zu, T., Bañez Coronel, M., Kelley, C. P., Redding-Ochoa, J., Nizamis, E., Melloni, A., Connors, T. R., Gaona, A., Thangaraju, K., Pletnikova, O., Clark, H. B., Davidson, B. L., Yachnis, A. T., … Ranum, L. P. W. (2025). CASP8 intronic expansion identified by poly-glycine-arginine pathology increases Alzheimer's disease risk. *Proceedings of the National Academy of Sciences of the United States of America, 122*(7), e2416885122. https://doi.org/10.1073/pnas.2416885122

Phuong, H. T., Tomas, R. F., Akmese, C., Mijares, A., Gerstin, I. M., Guo, S., Bell, L. R., Ellwood, R., Yegorova, S., Ng, S. K., Massey, G., Phillips, J., Melloni, A., Pletnikova, O., Lou, X., Clark, H. B., Troncoso, J. C., Hyman, B. T., Prokop, S., Ranum, L. P. W., … Nguyen, L. (2025). PolyGR-containing aggregates link with pathology and clinical features of Alzheimer's disease. *Acta neuropathologica, 150*(1), 49. https://doi.org/10.1007/s00401-025-02954-8